

Practical 2

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Malware Analysis Spring 2025 - P0x02

Executive Summary

From my analysis, I have concluded that sample2.exe is a raccoon / info stealer that connects back to a C&C server at IP address 255.245.168.192.in-addr.arpa, and tried to receive new files or instructions from “manillamemories(.com)” and also “telete(.in)”. This was done over a DNS connection.

The file, sample2.exe, loads a package of hidden code onto the computer likely through hidden .cab files. This will create another file called wotsuper1.exe, which has been observed to extract passwords, cookies, hardware/software information about the victim’s computer, and the victim’s IP address through two iplogger(.com) visits through Internet Explorer and an invasion of several web browsers seen in Process Monitor. The malware, after acquiring all this information and storing it in temporary files called passwords.txt and information.txt, then sends the data to an external C&C Server, as indicated by the WPAD settings being tampered with.

The reason I am sure that this is an information stealer is because of the information it is clearly gathering across the victim’s computer and the obvious exfiltrating of that said information back to the attacker.

Basic Static Analysis

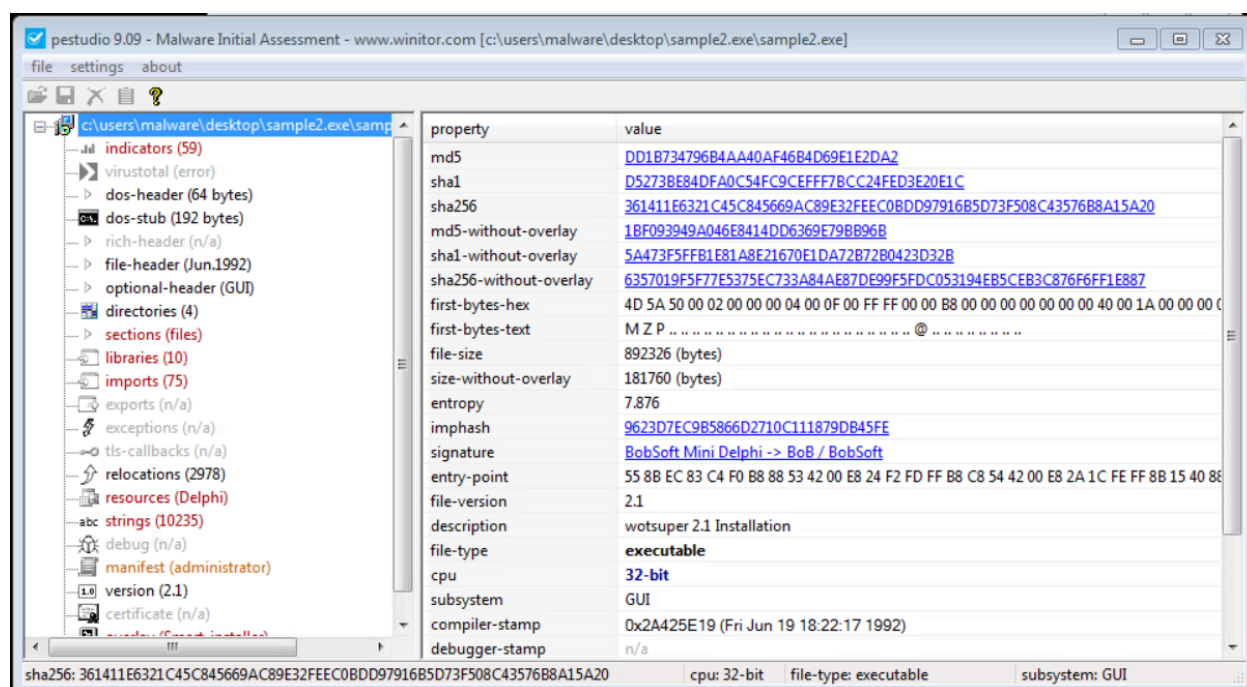


Figure 1: PEStudio Overview

From this screenshot of the apparent compilation date of the malware, which seems to be **June 19, 1992 at 6:22:17 pm**, although this is not the ACTUAL compilation date as hinted by the copyright messages found in the strings section that mention a copyright for 2001. In addition to this, we can observe that this program is a **Windows GUI program**. It is **32-bit compatible**. Its hashes, as it appears in PE Studio, are DD1B734796B4AA40AF46B4D69E1E2 DA2 (**MD5**), D5273BE84DFA0C54FC9CEFFF7BCC24FED3E20E1C (**SHA1**), and 361411E6321C45C845669AC89E32FEEC0BDD97916B5D73F508C43576B8A15A20 (**SHA256**). There are also versions of these hashes that are without the overlay, which this malware seems to be using to install itself onto the victim's PC, but this is just an assumption for now. The overlay section of this report will go into more detail on this.

[illegible]

Packing

From looking at the PEStudio Sections, the CODE section contains the entry point and is the only executable section. DATA, BSS, .idata, and .tls are all writable sections. There is clearly some packing going on here from the “Delphi, offset 0x...”. This is a known type of packing which is used to prevent malware from being detected or being found by reverse engineering. In addition to this, from the PEStudio overview, we can see the entropy is 7.876, which is greater than 7, which indicates this malware executable is packed.

name	CODE
md5	5E14E4EDE2E2215BC7D7283...
entropy	6.594
file-ratio (20.25%)	16.70 %
raw-address	0x00000400
raw-size (180736 bytes)	0x00024600 (148992 bytes)
virtual-address	0x00401000
virtual-size (183149 bytes)	0x000244CC (148684 bytes)
entry-point	0x00025468
characteristics	0x60000020

We can double check this in a few ways, one of which is looking at the file ratio. This ratio, in unpacked software is usually at or near 100%. This software is at 20.25%, which is a definitive sign of packing. If there is any doubt left however, there is one more telling sign that this malware is packed.

Referring back to *Figure 1*, the text corresponding to the “signature” section reads “BobSoft Mini Delphi”. From some research on the internet, I found that this is a known packer for malware that ensures that the malware is not in an analysis environment [1].

PE File Sections

The PE file sections are named **CODE**, **DATA**, **BSS**, **.idata**, **.tls**, **.rdata**, and **.rsrc**.

CODE is the section that contains the executable code, and is similar to **.text**. **DATA** is where *initialized* global and static variables are located. **BSS** is where *uninitialized* global and static variables are stored. **.idata** is where imported functions and libraries are stored that are used by the executable. **.tls** is short for “Thread Local Storage”, and stores some code that executes before the entry-point is executed, and is often used to set up the malware’s ideal environment to wreak havoc. **.rdata** contains constants that shouldn’t be modified during execution such as constant variables, string literals, etc. Lastly, **.rsrc** usually contains resources/data used by the OS or the program itself for managing resources.

These section headers may end up changing after performing unpacking, but these are the headers as they are before unpacking.

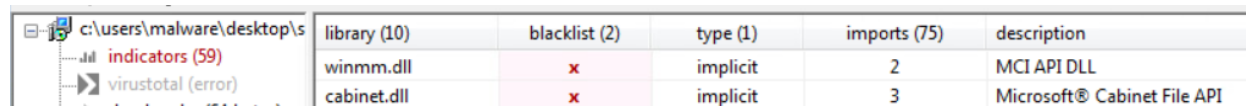
PE Function Imports

name (75)	group (9)	type (1)	ordinal (0)	blacklist (10)	anti-debug (0)	undocumented (0)	c
FDIDestroy	setup	implicit	-	x	-	-	
FDICopy	setup	implicit	-	x	-	-	
FDICreate	setup	implicit	-	x	-	-	
GetKeyboardType	keyboard-and-mouse	implicit	-	x	-	-	
SHGetFileInfoA	file	implicit	-	x	-	-	
GetCurrentThreadId	execution	implicit	-	x	-	-	
GetThreadLocale	execution	implicit	-	x	-	-	
RaiseException	exception-handling	implicit	-	x	-	-	
timeKillEvent	-	implicit	-	x	-	-	
timeSetEvent	-	implicit	-	x	-	-	
DeleteCriticalSection	synchronization	implicit	-	-	-	-	
LeaveCriticalSection	synchronization	implicit	-	-	-	-	
EnterCriticalSection	synchronization	implicit	-	-	-	-	
InitializeCriticalSection	synchronization	implicit	-	-	-	-	
RegQueryValueExA	registry	implicit	-	-	-	-	
RegOpenKeyExA	registry	implicit	-	-	-	-	
RegCloseKey	registry	implicit	-	-	-	-	
VirtualFree	memory	implicit	-	-	-	-	
VirtualAlloc	http://social.msdn.microsoft.com/Search/en-US/windows/desktop?query=VirtualFree	-	-	-	-	-	
LocalFree	memory	implicit	-	-	-	-	
LocalAlloc	memory	implicit	-	-	-	-	
WriteFile	file	implicit	-	-	-	-	
GetStartupInfoA	execution	implicit	-	-	-	-	
GetCommandLineA	execution	implicit	-	-	-	-	
ExitProcess	execution	implicit	-	-	-	-	
UnhandledExceptionFilter	exception-handling	implicit	-	-	-	-	
FreeLibrary	dynamic-library	implicit	-	-	-	-	
GetStdHandle	console	implicit	-	-	-	-	
GetVersion	-	implicit	-	-	-	-	

Figure 3: PESTudio Function Imports

From looking at the function imports, there are many functions that throw red flags. The first section of these that are suspicious are the **FDIDestroy**, **FDICopy**, and **FDICreate** functions which are used in the **FDI** (File Decompression Interface) system. These functions are used to extract files from

.CAB (cabinet) files, and are often found in installers and software updaters. .CAB files are hidden files that are used by malware authors to hide malicious code from anti-virus software. This is likely how the dangerous code is actually unraveled onto the victim's computer. To double check that this is actually going on, it's important to check that the cabinet.dll file is imported, as that basically confirms the interactions with .cab files. As seen in *Figure 3.1*, we identified that cabinet.dll was indeed imported.



library (10)	blacklist (2)	type (1)	imports (75)	description
winmm.dll	x	implicit	2	MCI API DLL
cabinet.dll	x	implicit	3	Microsoft® Cabinet File API

Figure 3.1: PEStudio Function Imports

The next suspicious function is GetKeyboardType. This function returns important information about the user's keyboard, including the type and subtype of keyboard (language type). This is often used by malware to detect whether or not it is being run in a virtual machine by detecting if the VM has just selected a generic keyboard. If so, the malware will often run in a passive or handicapped /non malicious state to throw off malware analysts. Another more unlikely way this is being used is to detect if the keyboard is from a specific region, and if so, it could adapt it's attack to that region, such as if the malware author only wants to attack those with a Chinese keyboard (basically a targeted cyber attack on China). Out of these two options, I suspect the former is probably the case.

Following this, the timeKillEvent and timeSetEvent functions are often used by malicious actors to create and stop events in their programs that act based on the state of the computer. For example, these may be used to create processes, and only exfiltrate data based on the state of the machine, such as when the computer is not a VM, or maybe when the computer is in idle for a set amount of time, although these may not be how they are used in this specific program.

The last of these functions that caught my eye were SHGetFileInfoA. This is used by malware authors to gather information about files on a victim's system.

Strings

When looking at the **strings**, some strings that seemed important were:

1. **Wotsuper 2.1 Installation, wotsuper@company .com, wotsuper.exe, and wotsuper. com** which seems to be the name, email, and website of the malware / malware author.
2. **[https://iplogger.org/1Ldta7 .html](https://iplogger.org/1Ldta7.html) and [https://iplogger.org/1smEq7 .html](https://iplogger.org/1smEq7.html)** (spaces added to make sure no one actually clicks this link), which is a known website that seems to log IP addresses of those who visit it and is often used by malicious actors to gather IP addresses of victims.
3. **GetKeyState**, which is the name of a function that returns the state of specific keyboard keys, which is an obvious sign of keylogging.
4. **Borland\Delphi\RTL**, which is the name of the packer, "Delphi".
5. **exitWindowsEx**, which is used to logout a user, turn off a computer, or restart a system. This is a very dangerous function for malware to be utilizing.

Other files that were imported include: **\sim.exe, ShellExecute, and RemoveDirectory** although I'm not sure of their significance at this point. There were also lots of files with random names

From doing some more research into **wotsuper 2.1**, I found that it is known malware [2]. It is used to steal data and return it to the attacker, although I of course will confirm this myself to be sure that it is indeed stealing data.

Exports

Next, I took a look at the function exports, but unlike Practical 1, there were no visible exports. This could be because of the Bobsoft / Delphi packing, but at this point I am unsure.

Manifest

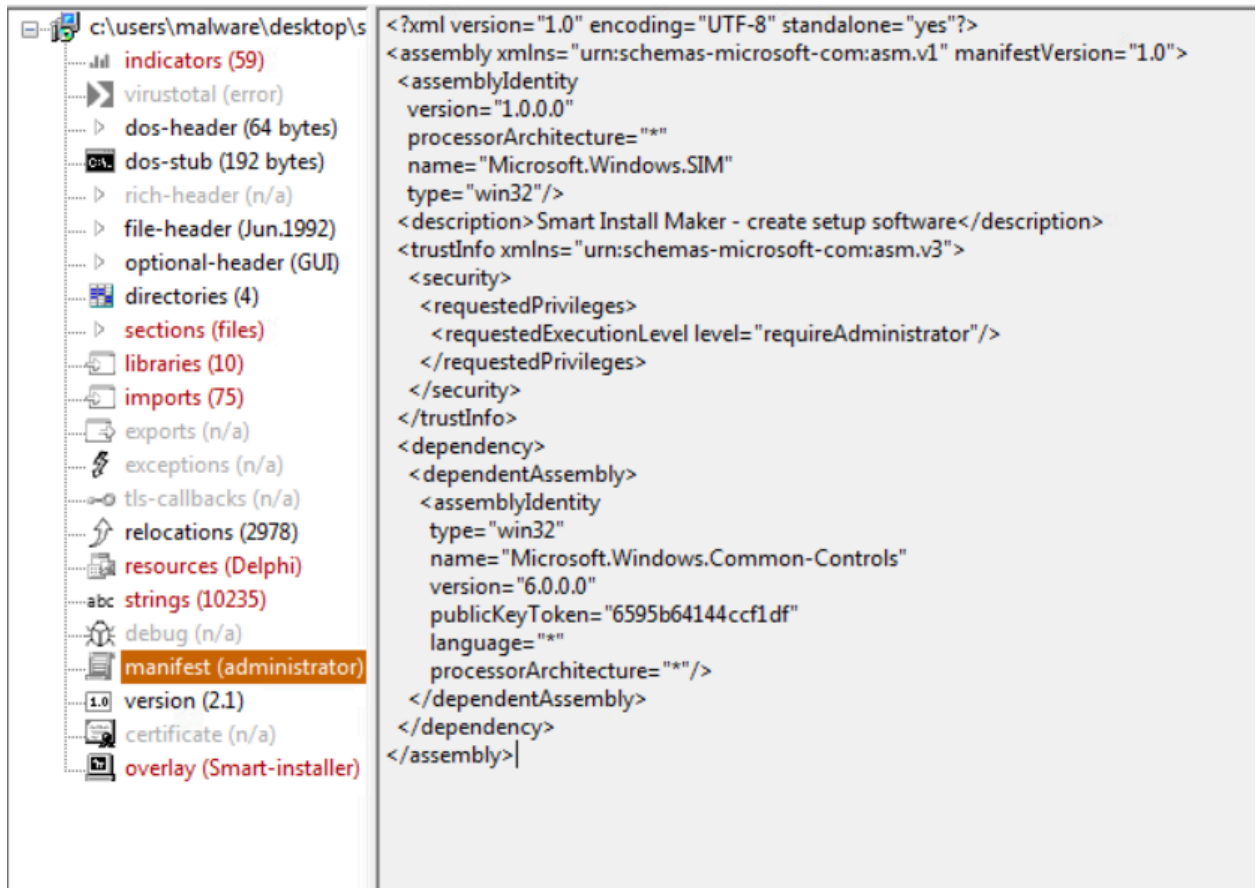


Figure 4: PESTudio Manifest section

In the manifest section, I discovered that the program requests administrative permissions on launch, according to the line `<requestedExecutionLevel level="requireAdministrator">`. Without running the program and knowing beforehand, I would say that this program will run and then immediately ask for admin permissions from the user.

Overlay

c:\users\malware\Desktop\	
indicators (59)	
virustotal (error)	
dos-header (64 bytes)	
dos-stub (192 bytes)	
rich-header (n/a)	
file-header (Jun.1992)	
optional-header (GUI)	
directories (4)	
sections (files)	
libraries (10)	
imports (75)	
exports (n/a)	
exceptions (n/a)	
tls-callbacks (n/a)	
relocations (2978)	
resources (Delphi)	
strings (10235)	
debug (n/a)	
manifest (administrator)	
version (2.1)	
certificate (n/a)	
overlay (Smart-installer)	

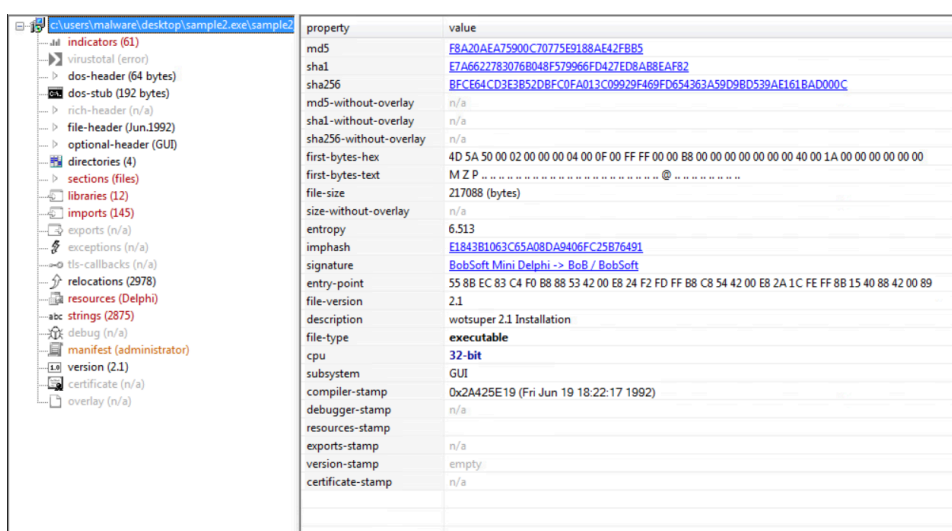
property	value
md5	86A248DC2BD0840445581D445640282F
sha1	54211CAFD97D19E282045D7CFFC58389FB24DEFB
sha256	0F4F4A5A03AD8144CB91257933AC8B246566CAF0C85664E7673E662F93FAB662
entropy	7.999
file-offset	0x0002C600
size	710566 (bytes)
signature	Smart-installer
first-bytes-hex	53 6D 61 72 74 20 49 6E 73 74 61 6C 6C 20 4D 61 6B 65 72 20 76 2E 20 35 2E 30 34 00 30 00 30 00
first-bytes-text	Smart Install Maker v. 5.04..0..0..
file-ratio	79.63 %

Figure 5: PEStudio Overlay Section

Finally, I took a look at the overlay section in PEStudio, and found that it takes up the other 79.63% of the file-ratio. This is extremely suspicious that this much extra code is in this program, and this is probably where most of the malicious code is stored. This is done so the code is less noticeable and less detectable to antivirus software and malware analysts.

Scylla / x32dbg Dynamic Analysis

Following the identification of packing found in the previous section, I needed a way to unpack this software to see all of the true imports and functions of the malware. Thankfully, within x32dbg, there is a plugin called Scylla that conveniently unpacks malware like this. I discovered that unless I unpack the software at the beginning of when CreateFileA is called, there were no changes to the imports. However, after finding the first instance of CreateFileA after running “bp CreateFileA” as a command in x32dbg, unpacking it using Scylla and then importing that Scylla rebuilt dump into PEStudio, there were 70 new imports. My working theory at this point is that this is from the unpacked code in the overlay section that has now moved over to the CODE section. This seems like a lower priority task for this assignment, since I already know it has been packed with Delphi.



property	value
md5	F8A20AE75900C70775E9188AE42F8B5
sha1	E7A662783076B048F57996FD47ED8A88EAFB2
sha256	BFC664CD3E3B52DBFC0FA013C09929F469FD654363A59D98D539AE161BAD000C
md5-without-overlay	n/a
sha1-without-overlay	n/a
sha256-without-overlay	n/a
first-bytes-hex	4D 5A 50 00 02 00 00 00 04 00 0F 00 FF FF 00 00 B8 00 00 00 00 00 00 00 40 00 1A 00 00 00 00 00
first-bytes-text	M Z P@
file-size	217088 (bytes)
size-without-overlay	n/a
entropy	6.513
imphash	E1843B1063C65A08DA9406FC25B76491
signature	BobSoft Mini Delphi -> BoB / BobSoft
entry-point	55 8B EC 83 C4 F0 B8 88 53 42 00 E8 24 F2 FD FF B8 C8 54 42 00 E8 2A 1C FE FF 8B 15 40 88 42 00 89
file-version	2.1
description	wotsuper 2.1 Installation
file-type	executable
cpu	32-bit
subsystem	GUI
compiler-stamp	0x2A425E19 (Fri Jun 19 18:22:17 1992)
debugger-stamp	n/a
resources-stamp	n/a
exports-stamp	n/a
version-stamp	empty
certificate-stamp	n/a

Figure 6: PEStudio Overview Post-Unpacking

name (145)	group (11)	type (1)	ordinal (0)	blacklist (14)	anti-debug (0)	undocumented (0)	deprecated (10)	library (12)
FDIDestroy	setup	implicit	-	x	-	-	-	cabinet.dll
FDICopy	setup	implicit	-	x	-	-	-	cabinet.dll
FDICreate	setup	implicit	-	x	-	-	-	cabinet.dll
GetKeyboardType	keyboard-and-mouse	implicit	-	x	-	-	-	user32.dll
SHGetFileInfoA	file	implicit	-	x	-	-	x	shell32.dll
GetCurrentThreadId	execution	implicit	-	x	-	-	-	kernel32.dll
GetThreadLocale	execution	implicit	-	x	-	-	-	kernel32.dll
RaiseException	exception-handling	implicit	-	x	-	-	-	kernel32.dll
RtlCaptureContext	exception-handling	implicit	-	x	-	-	-	ntdll.dll
RtlCaptureStackBackTrace	diagnostic	implicit	-	x	-	-	-	ntdll.dll
timeKillEvent	-	implicit	-	x	-	-	-	winmm.dll
timeSetEvent	-	implicit	-	x	-	-	-	winmm.dll
EnumCalendarInfoW	-	implicit	-	x	-	-	-	kernelbase.dll
SetFileApisToOEM	-	implicit	-	x	-	-	-	kernelbase.dll
EnumSystemLocalesEx	system-information	implicit	-	-	-	-	-	kernelbase.dll
EnumSystemLocalesW	system-information	implicit	-	-	-	-	-	kernelbase.dll
DeleteCriticalSection	synchronization	implicit	-	-	-	-	-	kernel32.dll
LeaveCriticalSection	synchronization	implicit	-	-	-	-	-	kernel32.dll
EnterCriticalSection	synchronization	implicit	-	-	-	-	-	kernel32.dll
InitializeCriticalSection	synchronization	implicit	-	-	-	-	-	kernel32.dll
RegQueryValueExA	registry	implicit	-	-	-	-	-	advapi32.dll
RegOpenKeyExA	registry	implicit	-	-	-	-	-	advapi32.dll
RegCloseKey	registry	implicit	-	-	-	-	-	advapi32.dll
VirtualFree	memory	implicit	-	-	-	-	-	kernel32.dll
VirtualAlloc	memory	implicit	-	-	-	-	-	kernel32.dll
LocalFree	memory	implicit	-	-	-	-	x	kernel32.dll
LocalAlloc	memory	implicit	-	-	-	-	x	kernel32.dll
GetStringTypeA	memory	implicit	-	-	-	-	x	kernelbase.dll
VirtualAllocExNuma	memory	implicit	-	-	-	-	-	kernelbase.dll
WriteFile	file	implicit	-	-	-	-	-	kernel32.dll

Figure 7: PEStudio Imports Post-Unpacking

After unpacking the imports, there were a few new imports that caught my eye, including **SetFileApisToOEM**, which is apparently used in obfuscation using OEM encoding. More important than that, however, was the **EnumSystemLocalesEx**, **EnumSystemLocalesW**, and **EnumCalendarInfoW**. These are functions that contain critical system information, like the region of the user. These are most likely used in the anti-VM methods mentioned previously and work most likely by checking if the region and language of the user are defaulted to some value VMs generally use. In that case, the malware may act differently if detected in a virtual machine environment.

General Dynamic Analysis

To start, here are the steps I took before starting:

Windows 7:

- 1) Loaded Regshot
- 2) Loaded Process Explorer
- 3) Loaded process monitor, cleared all previous processes

Remnux:

- 1) Loaded FakeDNS
- 2) Loaded Wireshark and began listening

Back to Windows 7:

- 1) Took first shot on Regshot
- 2) Ran the malware (nothing special, just an .exe)
- 3) Waited about 2 minutes then took the second shot and paused the other tools

Created / Modified Registry Keys/Files

```
-----
Keys added:5
HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Uninstall\wotsuper 2.1
HKU\S-1-5-21-4118134989-2631507447-873320884-1000\Software\Microsoft\Internet Explorer\Recovery\AdminActive
HKU\S-1-5-21-4118134989-2631507447-873320884-1000\Software\Microsoft\Windows\CurrentVersion\Internet Settings\Wpad\{66AB3F9F-831D-4B2B-8CCF-F88E03627596}
HKU\S-1-5-21-4118134989-2631507447-873320884-1000\Software\Microsoft\Windows\CurrentVersion\Internet Settings\Wpad\{66AB3F9F-831D-4B2B-8CCF-F88E03627596}
HKU\S-1-5-21-4118134989-2631507447-873320884-1000\Software\Microsoft\Windows\CurrentVersion\Internet Settings\Wpad\{66AB3F9F-831D-4B2B-8CCF-F88E03627596}\00-50-56-87-fc-06
-----
Values added:28
HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Uninstall\wotsuper 2.1\DisplayName: "wotsuper 2.1"
HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Uninstall\wotsuper 2.1\DisplayVersion: "2.1"
HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Uninstall\wotsuper 2.1\VersionMajor: 0x00000002
HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Uninstall\wotsuper 2.1\VersionMinor: 0x00000001
HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Uninstall\wotsuper 2.1\Publisher: "wotsuper"
HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Uninstall\wotsuper 2.1\DisplayIcon: "C:\Program Files\wotsuper\wotsuper\uninstall.exe"
HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Uninstall\wotsuper 2.1\UninstallString: "C:\Program Files\wotsuper\wotsuper\uninstall.exe"
HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Uninstall\wotsuper 2.1\URLInfoAbout: "http://www.wotsuper.com/"
HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Uninstall\wotsuper 2.1\HelpLink: "wotsuper@company.com"
HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Uninstall\wotsuper 2.1\InstallLocation: "C:\Program Files\wotsuper\wotsuper\"
HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Uninstall\wotsuper 2.1\InstallSource: "C:\Users\malware\Desktop\swag\sample2.exe\"
HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Uninstall\wotsuper 2.1\InstallDate: "20250328"
HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Uninstall\wotsuper 2.1\Language: 0x00000419
HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Uninstall\wotsuper 2.1\EstimatedSize: 0x00000044
HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Uninstall\wotsuper 2.1\NoModify: 0x00000001
HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Uninstall\wotsuper 2.1\NoRepair: 0x00000001
HKU\S-1-5-21-4118134989-2631507447-873320884-1000\Software\Microsoft\Internet Explorer\Recovery\AdminActive: 0x00000000
HKU\S-1-5-21-4118134989-2631507447-873320884-1000\Software\Microsoft\Internet Explorer\Recovery\AdminActive\{2620C5FE-0C2D-11F0-AA56-0050568781DB}: 0x00000000
HKU\S-1-5-21-4118134989-2631507447-873320884-1000\Software\Microsoft\Internet Explorer\Recovery\AdminActive\{2689828A-0C2D-11F0-AA56-0050568781DB}: 0x00000000
HKU\S-1-5-21-4118134989-2631507447-873320884-1000\Software\Microsoft\Windows\CurrentVersion\Internet Settings\Wpad\{66AB3F9F-831D-4B2B-8CCF-F88E03627596}\CountP1\href\znyjner\Qrfgxb
HKU\S-1-5-21-4118134989-2631507447-873320884-1000\Software\Microsoft\Windows\CurrentVersion\Internet Settings\Wpad\{66AB3F9F-831D-4B2B-8CCF-F88E03627596}\WpadDecisionReason: 0x00000001
HKU\S-1-5-21-4118134989-2631507447-873320884-1000\Software\Microsoft\Windows\CurrentVersion\Internet Settings\Wpad\{66AB3F9F-831D-4B2B-8CCF-F88E03627596}\WpadDecisionTime: 86 E1 D9 0F 3A A0 D8 01
HKU\S-1-5-21-4118134989-2631507447-873320884-1000\Software\Microsoft\Windows\CurrentVersion\Internet Settings\Wpad\{66AB3F9F-831D-4B2B-8CCF-F88E03627596}\WpadDecisionTime: 86 E1 D9
HKU\S-1-5-21-4118134989-2631507447-873320884-1000\Software\Microsoft\Windows\CurrentVersion\Internet Settings\Wpad\{66AB3F9F-831D-4B2B-8CCF-F88E03627596}\WpadDecision: 0x00000000
HKU\S-1-5-21-4118134989-2631507447-873320884-1000\Software\Microsoft\Windows\CurrentVersion\Internet Settings\Wpad\{66AB3F9F-831D-4B2B-8CCF-F88E03627596}\WpadNetworkName: "Network
HKU\S-1-5-21-4118134989-2631507447-873320884-1000\Software\Microsoft\Windows\CurrentVersion\Shell Extensions\cached\{17FE9752-0B5A-4665-84CD-569794602F5C} \{7F918580-CB92-43C5-80A9-
```

Figure 8: Regshot comparison results from running the malware

explorer.exe	< 0.01	31,820 K	46,056 K	2100 Windows Explorer	Microsoft Corporation
vmtoolsd.exe	< 0.01	4,660 K	8,828 K	2348 VMware Tools Core Service	VMware, Inc.
procexp.exe	< 0.01	14,236 K	21,848 K	1376 Sysinternals Process Explorer	Sysinternals - www.sysinter...
Procmon.exe		3,652 K	6,792 K	876 Process Monitor	Sysinternals - www.sysinter...
Procmon.exe		105,404 K	63,780 K	3040	
wotsuper.exe	< 0.01	51,272 K	8,416 K	3356	
wotsuper1.exe		3,892 K	9,228 K	1040	
regedit.exe		1,720 K	5,784 K	480	
ieexplore.exe		5,792 K	19,440 K	2568	
ieexplore.exe		11,048 K	27,480 K	1312	
notepad.exe		3,136 K	7,272 K	320 Notepad	Microsoft Corporation

Figure 10: Process Explorer

Now looking at what's going on live with Process Explorer, there are a few interesting executables being run. Now obviously **wotsuper.exe** and **wotsuper1.exe** are the malware, but it's also interesting that the malware left **regedit.exe** open, which launches the Windows Registry Editor. The two wotsuper executables are persisting and Internet Explorer (IE) is opened when the malware is executed. The malware opens two instances of IE, both opening iplogger(.com) ([https://iplogger\(.org\)/1Ldta7.html](https://iplogger(.org)/1Ldta7.html) and [https://iplogger\(.org\)/1smEq7.html](https://iplogger(.org)/1smEq7.html)). This is how the malware author gets the IP of the user, although it is odd why they open two instances of it.

7:34:1...	sample2.exe	1840	CreateFile	C:\Program Files\Internet Explorer\ieproxy.dll
7:34:1...	sample2.exe	1840	QueryBasicInfor...	C:\Program Files\Internet Explorer\ieproxy.dll
7:34:1...	sample2.exe	1840	CloseFile	C:\Program Files\Internet Explorer\ieproxy.dll
7:34:1...	sample2.exe	1840	CreateFile	C:\Program Files\Internet Explorer\ieproxy.dll
7:34:1...	sample2.exe	1840	CreateFileMapp...	C:\Program Files\Internet Explorer\ieproxy.dll
7:34:1...	sample2.exe	1840	CreateFileMapp...	C:\Program Files\Internet Explorer\ieproxy.dll
7:34:1...	sample2.exe	1840	CreateFileMapp...	C:\Program Files\Internet Explorer\ieproxy.dll
7:34:1...	sample2.exe	1840	Load Image	C:\Program Files\Internet Explorer\ieproxy.dll
7:34:1...	sample2.exe	1840	CloseFile	C:\Program Files\Internet Explorer\ieproxy.dll

Figure 10: Process Monitor

In Process Monitor, I wanted to double check and see if IE proxy settings were being directly affected, and as seen above, they are. This is a sign the attacker may be using a proxy server to redirect network traffic from the victim's computer.

7:34:1...	sample2.exe	1840	CreateFile	C:\Windows\System32\regedit.exe
7:34:1...	sample2.exe	1840	CreateFile	C:\Windows\system\regedit.exe
7:34:1...	sample2.exe	1840	CreateFile	C:\Windows\regedit.exe
7:34:1...	sample2.exe	1840	QueryBasicInfor...	C:\Windows\regedit.exe
7:34:1...	sample2.exe	1840	CloseFile	C:\Windows\regedit.exe

Figure 11.1: Process Monitor - regedit.exe being opened by malware

I also wanted to confirm that regedit was opened by the malware itself and not some windows process, and I confirmed it was opened by the malware as seen above.

7:34:1...	sample2.exe	1840	QueryBasicInfor...	C:\Program Files\wotsuper\wotsuper1.exe	SUCCESS	CreationTime: 7/8/2020 6:56:50 PM, L
7:34:1...	sample2.exe	1840	QueryBasicInfor...	C:\Program Files\wotsuper\wotsuper.exe	SUCCESS	CreationTime: 7/9/2020 12:28:04 AM, I

Figure 11.2: Process Monitor - Potential actual compilation/creation date of the malware

While snooping through process monitor, I noticed that the creation time of wotsuper1.exe and wotsuper.exe were 7/8/2020 and 7/9/2020, respectively. This differs from the believed 1992 date of compilation, and even if these are not the exact dates, it proves that the malware was created later than the apparent compiler stamp says.

7:34:1...	wotsuper1.exe	1040	CreateFile	C:\ProgramData\20WQ2KPKG09X5VJLQ8IEYAM8R\files\passwords.txt	SUCCESS
7:34:1...	wotsuper1.exe	1040	CloseFile	C:\ProgramData\20WQ2KPKG09X5VJLQ8IEYAM8R\files\passwords.txt	SUCCESS

Figure 11.3: Process Monitor - Initial Creation of "passwords.txt" info file

7:34:2...	wotsuper1.exe	1040	CreateFile	C:\ProgramData\20WQ2KPKG09X5VJLQ8IEYAM8R\files\passwords.txt	SUCCESS
7:34:2...	wotsuper1.exe	1040	QueryStandardI...	C:\ProgramData\20WQ2KPKG09X5VJLQ8IEYAM8R\files\passwords.txt	SUCCESS
7:34:2...	wotsuper1.exe	1040	CreateFile	C:\Users\malware\AppData\Local\Chromium\User Data\Local State	PATH NOT FOUND
7:34:2...	wotsuper1.exe	1040	CreateFile	C:\Users\malware\AppData\Local\Chromium\User Data\	PATH NOT FOUND
7:34:2...	wotsuper1.exe	1040	CloseFile	C:\ProgramData\20WQ2KPKG09X5VJLQ8IEYAM8R\files\passwords.txt	SUCCESS
7:34:2...	wotsuper1.exe	1040	CreateFile	C:\ProgramData\20WQ2KPKG09X5VJLQ8IEYAM8R\files\passwords.txt	SUCCESS
7:34:2...	wotsuper1.exe	1040	QueryStandardI...	C:\ProgramData\20WQ2KPKG09X5VJLQ8IEYAM8R\files\passwords.txt	SUCCESS
7:34:2...	wotsuper1.exe	1040	CreateFile	C:\Users\malware\AppData\Local\Kometa\User Data\Local State	PATH NOT FOUND
7:34:2...	wotsuper1.exe	1040	CreateFile	C:\Users\malware\AppData\Local\Kometa\User Data\	PATH NOT FOUND
7:34:2...	wotsuper1.exe	1040	CloseFile	C:\ProgramData\20WQ2KPKG09X5VJLQ8IEYAM8R\files\passwords.txt	SUCCESS
7:34:2...	wotsuper1.exe	1040	CreateFile	C:\ProgramData\20WQ2KPKG09X5VJLQ8IEYAM8R\files\passwords.txt	SUCCESS
7:34:2...	wotsuper1.exe	1040	QueryStandardI...	C:\ProgramData\20WQ2KPKG09X5VJLQ8IEYAM8R\files\passwords.txt	SUCCESS
7:34:2...	wotsuper1.exe	1040	CreateFile	C:\Users\malware\AppData\Local\Amigo\User Data\Local State	PATH NOT FOUND
7:34:2...	wotsuper1.exe	1040	CreateFile	C:\Users\malware\AppData\Local\Amigo\User Data\	PATH NOT FOUND
7:34:2...	wotsuper1.exe	1040	CloseFile	C:\ProgramData\20WQ2KPKG09X5VJLQ8IEYAM8R\files\passwords.txt	SUCCESS
7:34:2...	wotsuper1.exe	1040	CreateFile	C:\ProgramData\20WQ2KPKG09X5VJLQ8IEYAM8R\files\passwords.txt	SUCCESS
7:34:2...	wotsuper1.exe	1040	QueryStandardI...	C:\ProgramData\20WQ2KPKG09X5VJLQ8IEYAM8R\files\passwords.txt	SUCCESS
7:34:2...	wotsuper1.exe	1040	CreateFile	C:\Users\malware\AppData\Local\Torch\User Data\Local State	PATH NOT FOUND
7:34:2...	wotsuper1.exe	1040	CreateFile	C:\Users\malware\AppData\Local\Torch\User Data\	PATH NOT FOUND
7:34:2...	wotsuper1.exe	1040	CloseFile	C:\ProgramData\20WQ2KPKG09X5VJLQ8IEYAM8R\files\passwords.txt	SUCCESS
7:34:2...	wotsuper1.exe	1040	CreateFile	C:\ProgramData\20WQ2KPKG09X5VJLQ8IEYAM8R\files\passwords.txt	SUCCESS
7:34:2...	wotsuper1.exe	1040	QueryStandardI...	C:\ProgramData\20WQ2KPKG09X5VJLQ8IEYAM8R\files\passwords.txt	SUCCESS
7:34:2...	wotsuper1.exe	1040	CreateFile	C:\Users\malware\AppData\Local\Orbitum\User Data\Local State	PATH NOT FOUND
7:34:2...	wotsuper1.exe	1040	CreateFile	C:\Users\malware\AppData\Local\Orbitum\User Data\	PATH NOT FOUND

Figure 11.4: Process Monitor - Stealing of passwords from web-browsers (part 1)

7:34:2...	wotsuper1.exe	1040	CreateFile	C:\Users\malware\AppData\Roaming\Waterfox\Profiles\	PATH NOT FOUND
7:34:2...	wotsuper1.exe	1040	CreateFile	C:\ProgramData\nss3.dll	NAME NOT FOUND
7:34:2...	wotsuper1.exe	1040	CreateFile	C:\Users\malware\AppData\Roaming\Specxstudios\Cyberfox\Profiles\	PATH NOT FOUND
7:34:2...	wotsuper1.exe	1040	CreateFile	C:\ProgramData\nss3.dll	NAME NOT FOUND
7:34:2...	wotsuper1.exe	1040	CreateFile	C:\Users\malware\AppData\Roaming\NETGATE Technologies\BlackHawk\Profiles\	PATH NOT FOUND
7:34:2...	wotsuper1.exe	1040	CreateFile	C:\ProgramData\nss3.dll	NAME NOT FOUND
7:34:2...	wotsuper1.exe	1040	CreateFile	C:\Users\malware\AppData\Roaming\Mozilla\Firefox\Profiles\	PATH NOT FOUND
7:34:2...	wotsuper1.exe	1040	CreateFile	C:\ProgramData\nss3.dll	NAME NOT FOUND
7:34:2...	wotsuper1.exe	1040	CreateFile	C:\Users\malware\AppData\Roaming\K-Meleon	NAME NOT FOUND
7:34:2...	wotsuper1.exe	1040	CreateFile	C:\ProgramData\20WQ2KPKG09X5VJLQ8IEYAM8R\files\passwords.txt	SUCCESS
7:34:2...	wotsuper1.exe	1040	QueryStandardI...	C:\ProgramData\20WQ2KPKG09X5VJLQ8IEYAM8R\files\passwords.txt	SUCCESS
7:34:2...	wotsuper1.exe	1040	CreateFile	C:\Users\malware\AppData\Local\Google\Chrome\User Data\Local State	PATH NOT FOUND
7:34:2...	wotsuper1.exe	1040	CreateFile	C:\Users\malware\AppData\Local\Google\Chrome\User Data\	PATH NOT FOUND
7:34:2...	wotsuper1.exe	1040	CloseFile	C:\ProgramData\20WQ2KPKG09X5VJLQ8IEYAM8R\files\passwords.txt	SUCCESS
7:34:2...	wotsuper1.exe	1040	CreateFile	C:\ProgramData\20WQ2KPKG09X5VJLQ8IEYAM8R\files\passwords.txt	SUCCESS
7:34:2...	wotsuper1.exe	1040	QueryStandardI...	C:\ProgramData\20WQ2KPKG09X5VJLQ8IEYAM8R\files\passwords.txt	SUCCESS
7:34:2...	wotsuper1.exe	1040	CreateFile	C:\Users\malware\AppData\Local\Chromium\User Data\Local State	PATH NOT FOUND
7:34:2...	wotsuper1.exe	1040	CreateFile	C:\Users\malware\AppData\Local\Chromium\User Data\	PATH NOT FOUND
7:34:2...	wotsuper1.exe	1040	CloseFile	C:\ProgramData\20WQ2KPKG09X5VJLQ8IEYAM8R\files\passwords.txt	SUCCESS
7:34:2...	wotsuper1.exe	1040	CreateFile	C:\ProgramData\20WQ2KPKG09X5VJLQ8IEYAM8R\files\passwords.txt	SUCCESS
7:34:2...	wotsuper1.exe	1040	QueryStandardI...	C:\ProgramData\20WQ2KPKG09X5VJLQ8IEYAM8R\files\passwords.txt	SUCCESS
7:34:2...	wotsuper1.exe	1040	CreateFile	C:\Users\malware\AppData\Local\Kometa\User Data\Local State	PATH NOT FOUND
7:34:2...	wotsuper1.exe	1040	CreateFile	C:\Users\malware\AppData\Local\Kometa\User Data\	PATH NOT FOUND
7:34:2...	wotsuper1.exe	1040	CloseFile	C:\ProgramData\20WQ2KPKG09X5VJLQ8IEYAM8R\files\passwords.txt	SUCCESS

Figure 11.5: Process Monitor - Stealing of passwords from web-browsers (part 2)

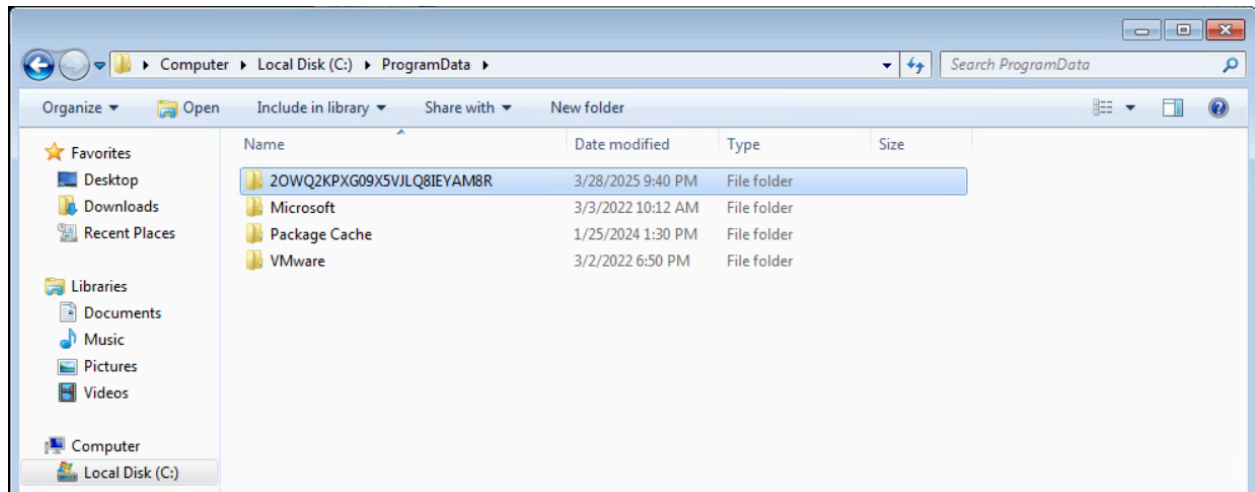


Figure 11.6: Process Monitor - Malicious directory “2OWQ2KPXG09X5VJLQ8IEYAM8R”

Looking through wotsuper1.exe’s processes now, I noticed something EXTREMELY odd. The program created a passwords.txt file that seems to be stealing all of the victims passwords. It is stored in this odd malware-created directory called “2OWQ2KPXG09X5VJLQ8IEYAM8R”, where it seems to be storing lots of these files it is doing work on. It originally created this directory soon after it was run by the original sample2.exe file. In the end this password file was deleted and probably attempted to be sent off to the attacker.

7:34:1...	wotsuper1.exe	1040	RegOpenKey	HKCU\Control Panel\Desktop\LanguageConfiguration	SUCCESS
7:34:1...	wotsuper1.exe	1040	RegEnumValue	HKCU\Control Panel\Desktop\LanguageConfiguration	NO MORE ENTRI...
7:34:1...	wotsuper1.exe	1040	RegCloseKey	HKCU\Control Panel\Desktop\LanguageConfiguration	SUCCESS

Figure 11.7: Process Monitor - Language Configuration being Accessed

I also noticed that the language configuration of the PC was being tracked.

7:34:1...	sample2.exe	1840	RegQueryValue	HKCU\Software\Microsoft\Windows\CurrentVersion\Explorer\User Shell Folders\Cookies	SUCCESS
7:34:1...	sample2.exe	1840	CreateFile	C:\Users\malware\AppData\Roaming\Microsoft\Windows\Cookies	SUCCESS
7:34:1...	sample2.exe	1840	QueryBasicInfor...	C:\Users\malware\AppData\Roaming\Microsoft\Windows\Cookies	SUCCESS
7:34:1...	sample2.exe	1840	CloseFile	C:\Users\malware\AppData\Roaming\Microsoft\Windows\Cookies	SUCCESS
7:34:1...	wotsuper1.exe	1040	RegQueryValue	HKCU\Software\Microsoft\Windows\CurrentVersion\Internet Settings\LeashLegacyCookies	NAME NOT FOUND
7:34:1...	wotsuper1.exe	1040	RegOpenKey	HKCU\Software\Microsoft\Windows\CurrentVersion\Internet Settings\5.0\Cache\Cookies	SUCCESS
7:34:1...	wotsuper1.exe	1040	RegQueryValue	HKCU\Software\Microsoft\Windows\CurrentVersion\Explorer\User Shell Folders\Cookies	SUCCESS
7:34:1...	wotsuper1.exe	1040	CreateFile	C:\Users\malware\AppData\Roaming\Microsoft\Windows\Cookies	NAME COLLISION
7:34:1...	wotsuper1.exe	1040	CreateFile	C:\Users\malware\AppData\Roaming\Microsoft\Windows\Cookies	SUCCESS
7:34:1...	wotsuper1.exe	1040	QueryBasicInfor...	C:\Users\malware\AppData\Roaming\Microsoft\Windows\Cookies	SUCCESS
7:34:1...	wotsuper1.exe	1040	CloseFile	C:\Users\malware\AppData\Roaming\Microsoft\Windows\Cookies	SUCCESS
7:34:1...	wotsuper1.exe	1040	CreateFile	C:\Users\malware\AppData\Roaming\Microsoft\Windows\Cookies	SUCCESS
7:34:1...	wotsuper1.exe	1040	QueryBasicInfor...	C:\Users\malware\AppData\Roaming\Microsoft\Windows\Cookies	SUCCESS
7:34:1...	wotsuper1.exe	1040	CloseFile	C:\Users\malware\AppData\Roaming\Microsoft\Windows\Cookies	SUCCESS
7:34:1...	wotsuper1.exe	1040	RegCloseKey	HKCU\Software\Microsoft\Windows\CurrentVersion\Internet Settings\5.0\Cache\Cookies	SUCCESS

Figure 11.8: Process Monitor - Cookies were being stolen from user

I also noticed that the user’s cookies were being accessed/stolen in both sample2.exe and wotsuper1.exe.

7:34:1...	wotsuper1.exe	1040	RegOpenKey	HKCU\Software\Microsoft\Internet Explorer\Security
7:34:1...	wotsuper1.exe	1040	RegQueryValue	HKCU\Software\Microsoft\Internet Explorer\Security\DisableSecuritySettingsCheck
7:34:1...	wotsuper1.exe	1040	RegCloseKey	HKCU\Software\Microsoft\Internet Explorer\Security
7:34:1...	wotsuper1.exe	1040	RegOpenKey	HKLM\SOFTWARE\Microsoft\Internet Explorer\Security
7:34:1...	wotsuper1.exe	1040	RegQueryValue	HKLM\SOFTWARE\Microsoft\Internet Explorer\Security\DisableSecuritySettingsCheck
7:34:1...	wotsuper1.exe	1040	RegCloseKey	HKLM\SOFTWARE\Microsoft\Internet Explorer\Security

Figure 11.9: Process Monitor - Security settings disabled in IE

The malware seems to be messing with the security settings in Internet Explorer. This makes sense since it is automatically opening Internet Explorer independent of the user.

7:34:2...	wotsuper1.exe	1040	RegCloseKey	HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Uninstall	SUCCESS
7:34:2...	wotsuper1.exe	1040	WriteFile	C:\ProgramData\20WQ2KPG09X5VJLQ8IYAM8R\files\information.txt	SUCCESS
7:34:2...	wotsuper1.exe	1040	CloseFile	C:\ProgramData\20WQ2KPG09X5VJLQ8IYAM8R\files\information.txt	SUCCESS

Figure 11.10: Process Monitor - information.txt file being created

Similar to the passwords.txt file, another file called information.txt was created by wotsuper1.exe. I cannot recover this file due to its deletion, however it most likely also stored some information that would be sent off to the attacker, probably relevant hardware/computer information about the victim's PC and network (or an IP perhaps?).

7:34:2...	wotsuper1.exe	1040	RegEnumKey	HKCU\Software\Microsoft\Windows\CurrentVersion\Internet Settings\Wpad\{66AB3F9F-831D-4B...	SUCCESS	Index: 0, Name: 00-50-56-87fc-06
7:34:2...	wotsuper1.exe	1040	RegCreateKey	HKCU\Software\Microsoft\Windows\CurrentVersion\Internet Settings\Wpad\00-50-56-87fc-06	SUCCESS	Desired Access: Set Value, Disposition: REG_OPENED_EXISTING_KEY
7:34:2...	wotsuper1.exe	1040	RegSetValue	HKCU\Software\Microsoft\Windows\CurrentVersion\Internet Settings\Wpad\00-50-56-87fc-06W...	SUCCESS	Type: REG_DWORD, Length: 4, Data: 1
7:34:2...	wotsuper1.exe	1040	RegSetValue	HKCU\Software\Microsoft\Windows\CurrentVersion\Internet Settings\Wpad\00-50-56-87fc-06W...	SUCCESS	Type: REG_BINARY, Length: 8, Data: 05 49 40 EF 39 A0 D8 01
7:34:2...	wotsuper1.exe	1040	RegSetValue	HKCU\Software\Microsoft\Windows\CurrentVersion\Internet Settings\Wpad\00-50-56-87fc-06W...	SUCCESS	Type: REG_DWORD, Length: 4, Data: 0
7:34:2...	wotsuper1.exe	1040	RegDeleteValue	HKCU\Software\Microsoft\Windows\CurrentVersion\Internet Settings\Wpad\00-50-56-87fc-06W...	SUCCESS	
7:34:2...	wotsuper1.exe	1040	RegCloseKey	HKCU\Software\Microsoft\Windows\CurrentVersion\Internet Settings\Wpad\00-50-56-87fc-06	SUCCESS	
7:34:2...	wotsuper1.exe	1040	RegEnumKey	HKCU\Software\Microsoft\Windows\CurrentVersion\Internet Settings\Wpad\{66AB3F9F-831D-4B...	NO MORE ENTRI...	Index: 1, Length: 288
7:34:2...	wotsuper1.exe	1040	RegOpenKey	HKCU\Software\Microsoft\Windows\CurrentVersion\Internet Settings\Wpad\{66AB3F9F-831D-4B...	SUCCESS	
7:34:2...	wotsuper1.exe	1040	RegCreateKey	HKCU\Software\Microsoft\Windows\CurrentVersion\Internet Settings\Wpad\00-50-56-87fc-06	SUCCESS	Desired Access: Set Value, Create Sub Key
7:34:2...	wotsuper1.exe	1040	RegQueryValue	HKCU\Software\Microsoft\Windows\CurrentVersion\Internet Settings\Wpad\00-50-56-87fc-06W...	SUCCESS	Desired Access: Query Value, Set Value, Disposition: REG_OPENED_EXISTING_KEY
7:34:2...	wotsuper1.exe	1040	RegQueryValue	HKCU\Software\Microsoft\Windows\CurrentVersion\Internet Settings\Wpad\00-50-56-87fc-06W...	SUCCESS	Type: REG_DWORD, Length: 4, Data: 0
7:34:2...	wotsuper1.exe	1040	RegQueryValue	HKCU\Software\Microsoft\Windows\CurrentVersion\Internet Settings\Wpad\00-50-56-87fc-06W...	SUCCESS	Type: REG_BINARY, Length: 8, Data: 05 49 40 EF 39 A0 D8 01
7:34:2...	wotsuper1.exe	1040	RegQueryValue	HKCU\Software\Microsoft\Windows\CurrentVersion\Internet Settings\Wpad\00-50-56-87fc-06W...	SUCCESS	Type: REG_DWORD, Length: 4, Data: 1
7:34:2...	wotsuper1.exe	1040	RegCreateKey	HKCU\Software\Microsoft\Windows\CurrentVersion\Internet Settings\Wpad\{66AB3F9F-831D-4B...	SUCCESS	Desired Access: Read, Disposition: REG_OPENED_EXISTING_KEY
7:34:2...	wotsuper1.exe	1040	RegSetValue	HKCU\Software\Microsoft\Windows\CurrentVersion\Internet Settings\Wpad\00-50-56-87fc-06W...	SUCCESS	Type: REG_DWORD, Length: 4, Data: 1
7:34:2...	wotsuper1.exe	1040	RegSetValue	HKCU\Software\Microsoft\Windows\CurrentVersion\Internet Settings\Wpad\00-50-56-87fc-06W...	SUCCESS	Type: REG_BINARY, Length: 8, Data: 05 49 40 EF 39 A0 D8 01
7:34:2...	wotsuper1.exe	1040	RegSetValue	HKCU\Software\Microsoft\Windows\CurrentVersion\Internet Settings\Wpad\00-50-56-87fc-06W...	SUCCESS	Type: REG_DWORD, Length: 4, Data: 0
7:34:2...	wotsuper1.exe	1040	RegDeleteValue	HKCU\Software\Microsoft\Windows\CurrentVersion\Internet Settings\Wpad\00-50-56-87fc-06W...	NAME NOT FOUND	Length: 144
7:34:2...	wotsuper1.exe	1040	RegCloseKey	HKCU\Software\Microsoft\Windows\CurrentVersion\Internet Settings\Wpad\{66AB3F9F-831D-4B...	SUCCESS	
7:34:2...	wotsuper1.exe	1040	RegCloseKey	HKCU\Software\Microsoft\Windows\CurrentVersion\Internet Settings\Wpad\00-50-56-87fc-06	SUCCESS	

Figure 11.11: Process Monitor - WPAD settings being tampered with

Lastly, WPAD settings were being tampered with at the end of this executable process. It is likely exfiltrating the gathered information and passwords to the attacker at this point.

(Note: I also looked into wotsuper.exe on Process Monitor, although I didn't find anything that stuck out there that wasn't already apparent or done in the other files.)

Uninstall	3/29/2025 12:46 AM	Application	98 KB
Uninstall	3/29/2025 12:46 AM	Configuration sett...	3 KB
wotsuper	7/9/2020 12:28 AM	Application	450 KB
wotsuper1	7/8/2020 6:56 PM	Application	545 KB

Figure 12: Uninstall files created by wotsuper1.exe before running Uninstall application

Name	Date modified	Type	Size
wotsuper	7/9/2020 12:28 AM	Application	450 KB
wotsuper1	7/8/2020 6:56 PM	Application	545 KB

Figure 12.1: Uninstall files created by wotsuper1.exe after running Uninstall application

Before moving on to network findings, I wanted to mention that there was a file created by wotsuper1.exe called wotsuper, and inside was an Uninstall application. I decided to run this application and did not see any changes besides the deletion of the Uninstall application and configuration file. This is probably meant to be used by the malware itself it wants to properly uninstall, and does not work correctly if used by the victim.

106	38.944262044	192.168.245.133	192.168.245.133	DNS	79	Standard query 0x3105 A manillamemories.com
107	38.944885993	192.168.245.133	192.168.245.133	DNS	95	Standard query response 0x3105 A manillamemories.com A 192.168...
108	39.081048765	192.168.245.133	192.168.245.133	DNS	72	Standard query 0x2351 A iplogger.org
109	39.081633071	192.168.245.133	192.168.245.133	DNS	88	Standard query response 0x2351 A iplogger.org A 192.168.245.133
112	39.175686806	192.168.245.133	192.168.245.133	DNS	69	Standard query 0x8ff6 A telete.in
113	39.176014240	192.168.245.133	192.168.245.133	DNS	85	Standard query response 0x8ff6 A telete.in A 192.168.245.133
128	40.153510534	192.168.245.133	192.168.245.133	DNS	88	Standard query 0xa645 PTR 255.245.168.192.in-addr.arpa
129	40.154200347	192.168.245.133	192.168.245.133	DNS	104	Standard query response 0xa645 PTR 255.245.168.192.in-addr.arpa...

Figure 13: Wireshark capturing DNS results from running the malware

Running the malware resulted in a DNS lookup of an IP address: **255.245.168.192.in-addr.arpa**, which seems to be where some data is being sent. That seems to be a machine the service is attempting to contact by IP. However, unlike Practical 1, there are more issues. The malware attempts to go to a website called “**manillamemories(.com)**” and also “**telete(.in)**”. I was curious about what VirusTotal had to say about these sites, and both were flagged as malicious. After doing some more research, telete(.in) is a known website used in Racoon malware, according to the CSK (Cyber Swachhta Kendra, an Indian government malware investigation center) [3]. The other site I could not find any specific information on, but both are most likely involved in some sort of C&C server situation.

```
fakedns[INFO]: Response: manillamemories.com -> 192.168.245.133
fakedns[INFO]: Response: iplogger.org -> 192.168.245.133
fakedns[INFO]: Response: telete.in -> 192.168.245.133
fakedns[INFO]: Response: 255.245.168.192.in-addr.arpa -> 192.168.245.133
```

Figure 14: FakeDNS capturing DNS results from running the malware

Running FakeDNS seemed to give a similar result, and this seems to be an exfiltration system used to send data to a server that the malicious actors control (probably a CnC server). In addition to this, one of these sites must be where the malware can retrieve new instructions and files.

Indicators of Compromise

The main host-based indicators I observed were the changing of the WPAD keys to direct network traffic to an outside IP address, as well as wotsuper.exe and wotsuper1.exe continuously running long after execution. In addition to this, the malware is using code stored in .cab files to carry out its attack on the victim. This .cab file code hides what the malware is really trying to do, which seems to be exfiltrating the victim's passwords, IP address, and computer information.

The main network-based indicators I observed were the suspicious DNS requests found in Wireshark and FakeDNS that were trying to use Reverse DNS lookups to send data back to the attacker's server and possibly retrieve new files from those sites or IPs as well. The IP that was responsible was **255.245.168.192.in-addr.arpa**, and the involved websites were “**manillamemories(.com)**” and “**telete(.in)**”.

A terminal window with a dark background and light-colored text. The prompt is 'remnux@remnux:~/Desktop\$'. The first command is 'yara -r p2.yar sample2.exe', followed by the output 'p2 sample2.exe'. The second command is 'yara -r p2.yar PMA\ Labs/', followed by the output 'remnux@remnux:~/Desktop\$' on a new line, indicating no match was found.

```
remnux@remnux:~/Desktop$ yara -r p2.yar sample2.exe
p2 sample2.exe
remnux@remnux:~/Desktop$ yara -r p2.yar PMA\ Labs/
remnux@remnux:~/Desktop$
```

Figure 15: Yara rule successfully identifying sample2.exe and not anything from the PMA Labs

Here is the YARA rule I ended up using to identify sample2.exe:

```
rule p2
{
    strings:
        $email = "wotsuper@company.com"
        $malName = "wotsuper.exe"

    condition:
        $email and $malName
}
```

Although simple, it does not trigger the PMA binaries, and would successfully identify this piece of malware.

Appendix

[1] *Bobsoft Mini Delphi Packer - UNPROTECT Project*. Unprotect Techniques. (n.d.).

<https://unprotect.it/technique/bobsoft-mini-delphi-packer/>

[2] *Malware analysis wotsuper.exe malicious activity: ANY.RUN - malware sandbox online*. Malware analysis wotsuper.exe Malicious activity | ANY.RUN - Malware Sandbox Online. (n.d.).

<https://any.run/report/156995132bf65af33100b4801353dff6c6d4ae416d2184ba67fac6839c3f134c/6dfb1bf5-59b2-41c2-9e60-0b8b41944952>

[3] *Raccoon Infostealer malware*. Cyber Swachhta Kendra: Raccoon infostealer malware. (n.d.).

https://www.csk.gov.in/alerts/Raccoon_stealer.html